

AP CSP | Chapter 3: Study Goal Tracker

Part A | Understand, plan and implement

Name: _____ Partner: _____ Date: _____

Purpose: give feedback on three study sessions. Valid inputs: exactly three whole-number minute values, each from 0 to 180.

Requirements

For each session, display "Goal met" when minutes ≥ 30 and add one to goalHits; otherwise display "Keep going". After all three sessions, display goalHits, then "On track" if goalHits ≥ 2 , or "Try again" otherwise.

1. Identify the input, the two kinds of output, and the intended user.

2. Plan the program. State what must happen before, inside and after the loop.

Phase	Actions and variables
Before the loop	
Inside each repetition	
After the loop	

3. Write your pseudocode here or attach your saved code. Include assignment, selection and repetition.

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Part B | Trace, test and revise

4. Before running, predict the goal count and final feedback for each test. Then record the actual result.

Three inputs	Expected count / final text	Actual count / final text
29, 30, 31		
0, 0, 0		
30, 30, 30		
Your test:		

5. Trace the boundary test 29, 30, 31. Start with goalHits = 0.

Input minutes	Condition ≥ 30	Message this time	goalHits after
29			
30			
31			

6. Explain why the counter is initialized outside the loop and what goes wrong if it is reset inside.

7. Describe one bug or revision. Identify the earliest incorrect step and the test evidence after your change.

8. Extend the requirement: change the goal to 45 minutes. Give a new pair of boundary tests and the expected feedback.
